

❖ Domain Name System (DNS):

"DNS (Domain Name System) is a protocol that converts a website name (domain name) into its IP address. When we enter `www.amazon.com` in the browser, the browser first sends a request to the DNS server asking for the IP address of that website. The DNS server returns the correct IP address, and then the browser uses that IP address to connect to the web server and open the website."

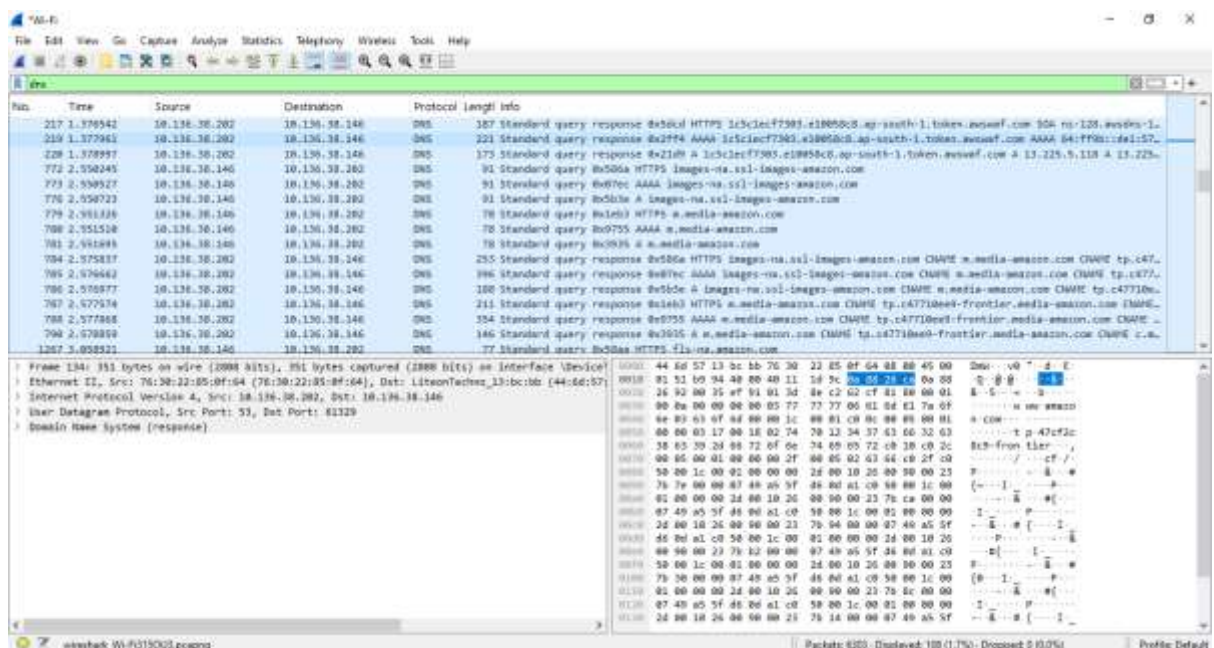
"DNS (Domain Name System) is a protocol that translates a domain name into an IP address so that computers can communicate with the correct server."

- Start Wireshark packet capture.
- Open your browser and visit <https://www.amazon.com/>

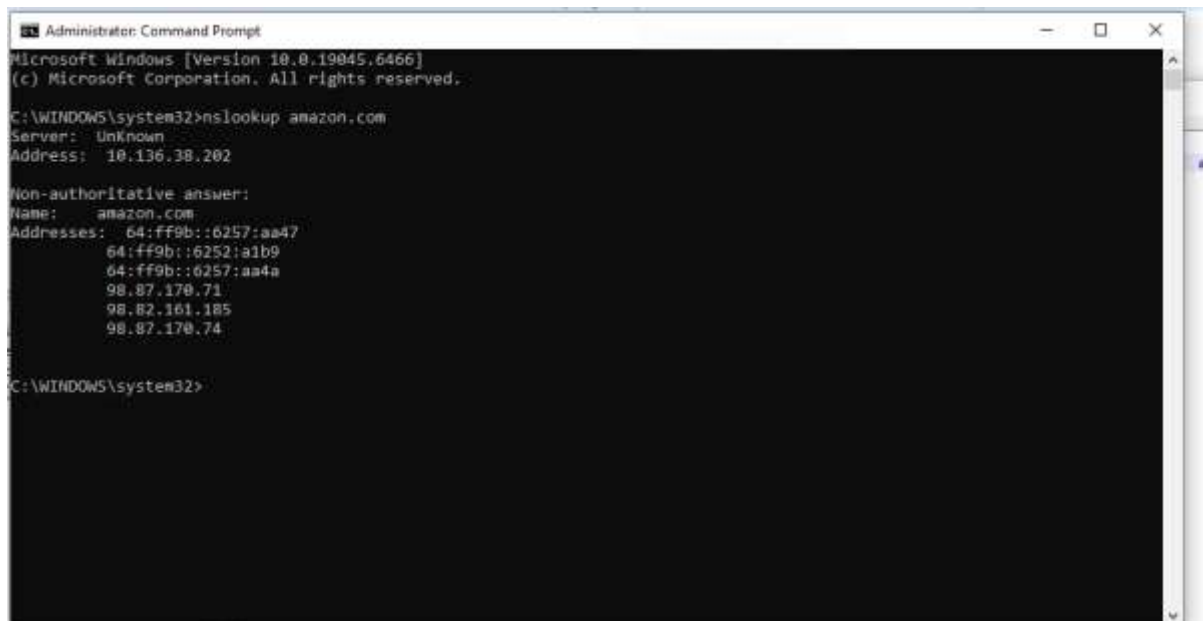
This generates DNS packets because the browser first requests the IP address of `amazon` from the DNS server.

- After the website opens successfully, **stop the Wireshark packet capture.**

1. Use the filtering functionality of Wireshark by typing "dns" in the filtering field, new window will appear:



3. At Command prompt ; Nslookup <https://www.amazon.com/>



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>nslookup amazon.com
Server: Unknown
Address: 10.136.38.202

Non-authoritative answer:
Name:    amazon.com
Addresses: 64:ff9b::6257:aa47
           64:ff9b::6252:a1b9
           64:ff9b::6257:aa4a
           98.87.170.71
           98.82.161.185
           98.87.170.74

C:\WINDOWS\system32>
```

4. Locate the DNS Query and Response message, Are they sent on UDP or TCP

ANS: The DNS Query and DNS Response are sent using the UDP protocol (Protocol 17).

The image shows a Wireshark packet capture of a DNS query and response. The packet list on the left shows a query (No. 42) and a response (No. 135). The packet details pane on the right shows the structure of the DNS query (No. 42). The packet bytes pane on the right shows the raw data of the DNS query.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
42	0.911853	10.136.38.140	10.136.38.202	DNS	77	Standard query 0x0022 AAAA fonts.gstatic.com
43	0.912191	10.136.38.140	10.136.38.202	DNS	77	Standard query 0x0022 A fonts.gstatic.com
44	0.913278	10.136.38.202	10.136.38.140	DNS	180	Standard query response 0x0022 AAAA fonts.gstatic.com AAAA 2884-6898-4089:1818:15e
45	0.913434	10.136.38.202	10.136.38.140	DNS	93	Standard query response 0x0022 A fonts.gstatic.com A 142.251.186.64
47	0.998787	10.136.38.202	10.136.38.140	DNS	182	Standard query response 0x0020 HTTPS fonts.gstatic.com HTTPS
124	1.209551	10.136.38.140	10.136.38.202	DNS	74	Standard query 0x0061 HTTPS www.amazon.com
125	1.209812	10.136.38.140	10.136.38.202	DNS	74	Standard query 0x00cf AAAA www.amazon.com
126	1.209278	10.136.38.140	10.136.38.202	DNS	74	Standard query 0x006d A www.amazon.com
130	1.219741	10.136.38.140	10.136.38.202	DNS	95	Standard query 0x0747 HTTPS optimizationguide-pa.googleapis.com
131	1.219129	10.136.38.140	10.136.38.202	DNS	95	Standard query 0x058c AAAA optimizationguide-pa.googleapis.com
132	1.219489	10.136.38.140	10.136.38.202	DNS	95	Standard query 0x009f A optimizationguide-pa.googleapis.com
133	1.220599	10.136.38.202	10.136.38.140	DNS	281	Standard query response 0x058c1 HTTPS www.amazon.com CNAME tp-47cf28c9-frontier.amazon.com CNAME cf-47cf28c9-
134	1.222336	10.136.38.202	10.136.38.140	DNS	161	Standard query response 0x02cf A www.amazon.com CNAME tp-47cf28c9-frontier.amazon.com CNAME cf-47cf28c9-
135	1.222953	10.136.38.202	10.136.38.140	DNS	145	Standard query response 0x006d A www.amazon.com CNAME tp-47cf28c9-frontier.amazon.com CNAME cf-47cf28c9-
137	1.229168	10.136.38.202	10.136.38.140	DNS	152	Standard query response 0x0747 HTTPS optimizationguide-pa.googleapis.com 504 nsl.google.cn
138	1.230725	10.136.38.202	10.136.38.140	DNS	219	Standard query response 0x058c AAAA optimizationguide-pa.googleapis.com AAAA 2001:4800:4840:000::AAAA 2001-

Packet Details (No. 42):

- Internet Protocol Version 4, Src: 10.136.38.140, Dst: 10.136.38.202
- 8000 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 68
- Identification: 0xb06c (57452)
- 0000 = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to live: 32
- Protocol: UDP (17)
- Header checksum: 0xf708 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.136.38.140
- Destination Address: 10.136.38.202
- [Stream Index: 0]
- User Datagram Protocol, Src Port: 58748, Dst Port: 55
- Domain Name System (query)

Packet Bytes:

0000 74 30 22 05 0f c4 44 6d 17 13 00 00 00 45 00 50" rDN 1... E

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134	1.222336	10.136.38.202	10.136.38.140	DNS	161	Standard query response 0x02cf A www.amazon.com CNAME tp-47cf28c9-frontier.amazon.com CNAME cf-47cf28c9-
135	1.222953	10.136.38.202	10.136.38.140	DNS	145	Standard query response 0x006d A www.amazon.com CNAME tp-47cf28c9-frontier.amazon.com CNAME cf-47cf28c9-
137	1.229168	10.136.38.202	10.136.38.140	DNS	152	Standard query response 0x0747 HTTPS optimizationguide-pa.googleapis.com 504 nsl.google.cn
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- Internet Protocol Version 4, Src: 10.136.38.140, Dst: 10.136.38.202
- 8000 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 68
- Identification: 0xb06c (57452)
- 0000 = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to live: 32
- Protocol: UDP (17)
- Header checksum: 0xf708 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.136.38.140
- Destination Address: 10.136.38.202
- [Stream Index: 0]
- User Datagram Protocol, Src Port: 58748, Dst Port: 55
- Domain Name System (query)

Packet Bytes:

0000 44 6d 17 13 00 00 00 45 00 50" rDN 1... E

5. What is the destination port for the DNS query message ? What is the source port of of DNS response message?

The destination port for the DNS Query message is Port 53, and the source port for the DNS Response message is also Port 53.

The screenshot shows a Wireshark packet capture of a DNS query message. The packet list on the left shows a series of DNS queries and responses. The selected packet is a DNS query from 10.136.38.202 to 10.136.38.202, protocol DNS, length 100 bytes. The packet details pane shows the following structure:

- Frame 124: 74 bytes on wire (592 bits), 76 bytes captured (592 bits) on interface DeviceNPF
- Ethernet II, Src: LitsanTechno_13:bc:bb (44:6d:57:13:bc:bb), Dst: 76:38:22:85:0f:64 (76:38:22:85:0f:64)
- Internet Protocol Version 4, Src: 10.136.38.146, Dst: 10.136.38.202
- User Datagram Protocol, Src Port: 56746, Dst Port: 53
- Source Port: 56746
- Destination Port: 53
- Length: 80
- Checksum: 0x0000 (Unverified)
- [Checksum Status: Unverified]
- [Stream Index: 8]
- [Stream Racket Number: 1]
- [Timestamps]
- UDP payload (32 bytes)
- Domain Name System (query)

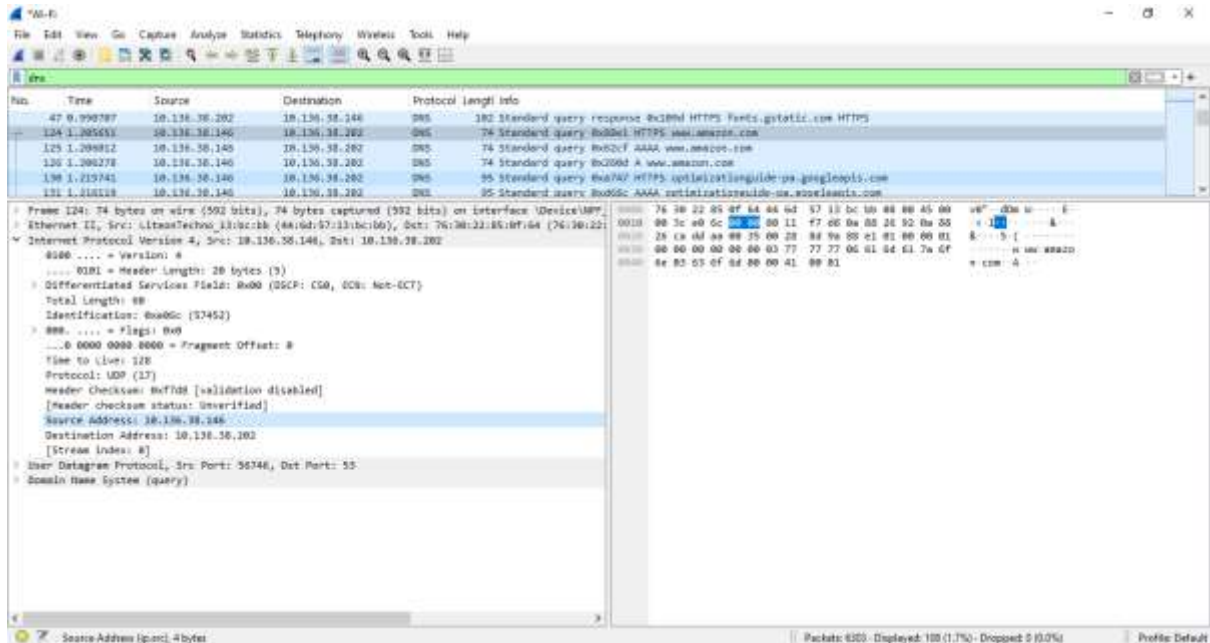
The packet bytes pane shows the raw data of the packet, including the DNS query structure.

The screenshot shows a Wireshark packet capture of a DNS response message. The packet list on the left shows a series of DNS queries and responses. The selected packet is a DNS response from 10.136.38.202 to 10.136.38.202, protocol DNS, length 280 bytes. The packet details pane shows the following structure:

- Frame 133: 280 bytes on wire (2240 bits), 280 bytes captured (2240 bits) on interface DeviceNPF
- Ethernet II, Src: 76:38:22:85:0f:64 (76:38:22:85:0f:64), Dst: LitsanTechno_13:bc:bb (44:6d:57:13:bc:bb)
- Internet Protocol Version 4, Src: 10.136.38.202, Dst: 10.136.38.146
- User Datagram Protocol, Src Port: 53, Dst Port: 56746
- Source Port: 53
- Destination Port: 56746
- Length: 167
- Checksum: 0x3524 (Unverified)
- [Checksum Status: Unverified]
- [Stream Index: 8]
- [Stream Racket Number: 2]
- [Timestamps]
- UDP payload (158 bytes)
- Domain Name System (response)

The packet bytes pane shows the raw data of the packet, including the DNS response structure.

6. What is the source IP address of the DNS Query message? Is it the IP address of your computer? Verify using `ipconfig`. Are these two IP address the same?



```
C:\WINDOWS\system32\ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

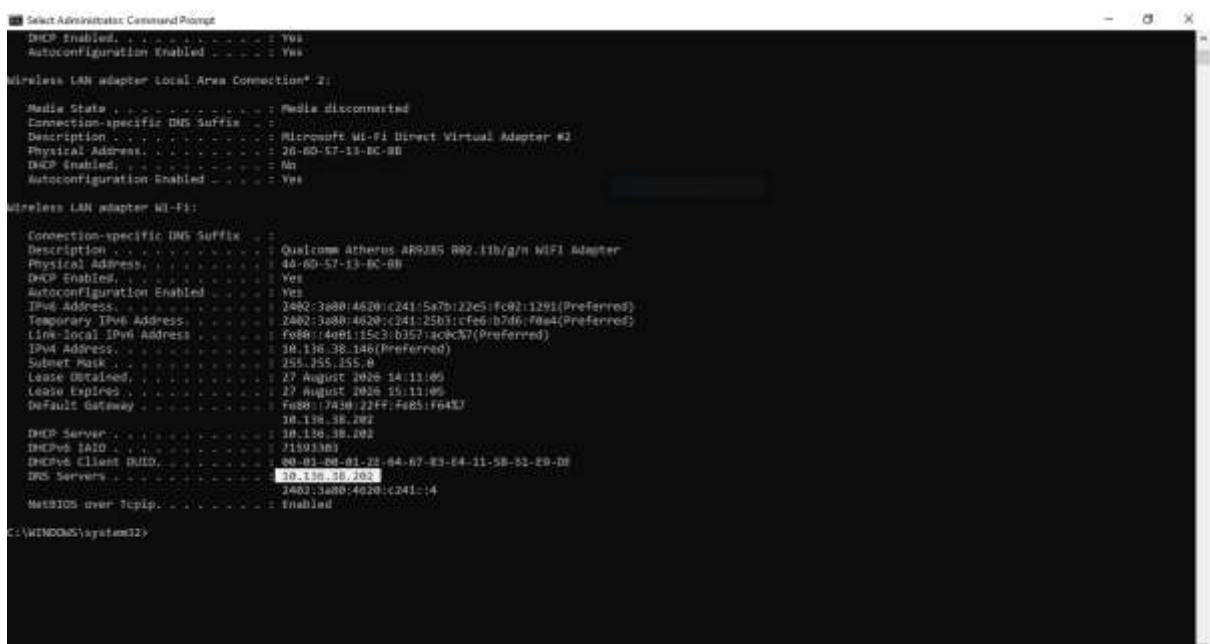
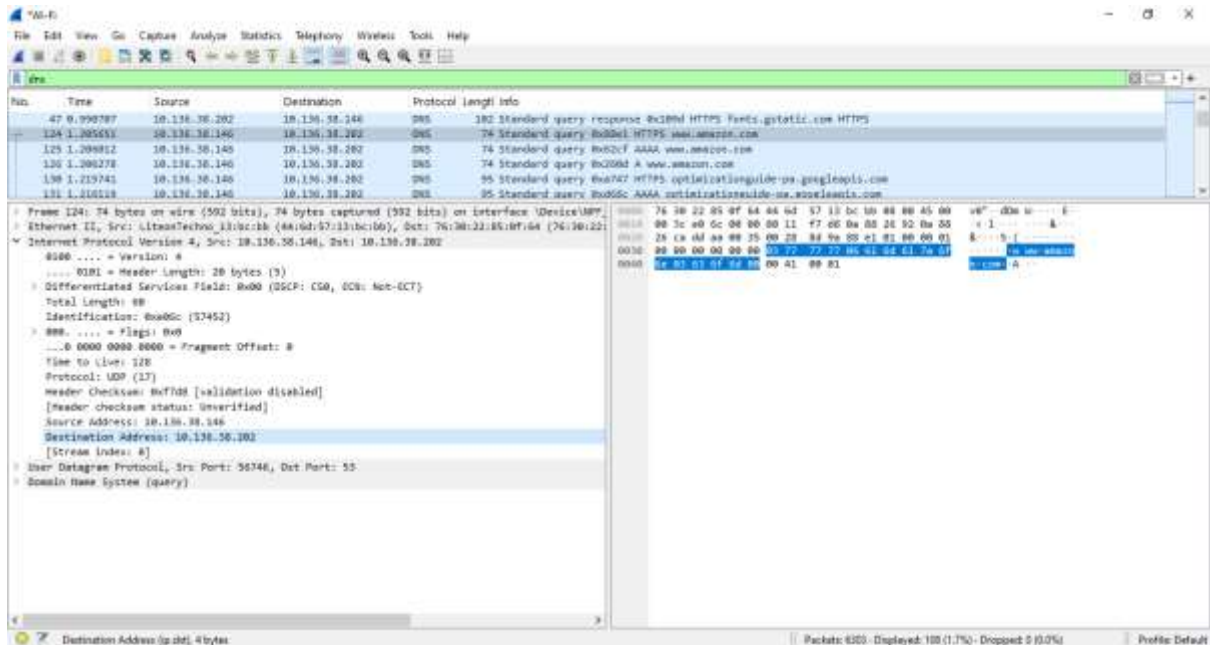
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2402:1a00:4000:f38a:8191:cd21:cbfa:7705
    Temporary IPv6 Address. . . . . : 2402:1a00:4000:f38a:f25e:2061:3426:c580
    Link-local IPv6 Address . . . . . : fe80::4a01:15:1b057:ac8c%7
    IPv4 Address. . . . . : 10.130.10.140
    Subnet Mask . . . . . : 255.255.155.0
    Default Gateway . . . . . : fe80::9496:2aff:fe01:f643%3
    10.130.10.202

C:\WINDOWS\system32>
```

7. What is the destination IP address of the DNS Query message? Is it the IP address of your local DNS server? Verify using `ipconfig`. Are these two IP address the same?



8. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?

Types: Https

Ans: 0

The image shows a Wireshark packet capture of a network traffic. The top pane displays a list of packets. Packet 124 is selected, showing a DNS query for www.amazon.com. The middle pane shows the details of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet 124: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 'Device\NPF...'

Ethernet II, Src: LltasTechno_12:35:bb (AA:6D:57:12:35:bb), Dst: 78:30:22:85:0F:64 (78:30:22:85:0F:64)

Source: LltasTechno_12:35:bb (AA:6D:57:12:35:bb)

Type: IPv4 (0x0008)

[Stream Index: 0]

Internet Protocol Version 4, Src: 10.136.38.140, Dst: 10.136.38.202

User Datagram Protocol, Src Port: 54746, Dst Port: 53

Domain Name System (query)

Transaction ID: 0x0001

Flags: 0x0100 Standard query

Questions: 1

Answer RRs: 0

Authority RRs: 0

Additional RRs: 0

Queries

www.amazon.com type HTTPS, class IN

Name: www.amazon.com

[Name Length: 14]

[Label Count: 3]

Type: HTTPS (65) (HTTPS Specific Service Endpoints)

Class: IN (0x0001)

[Response To: 123]

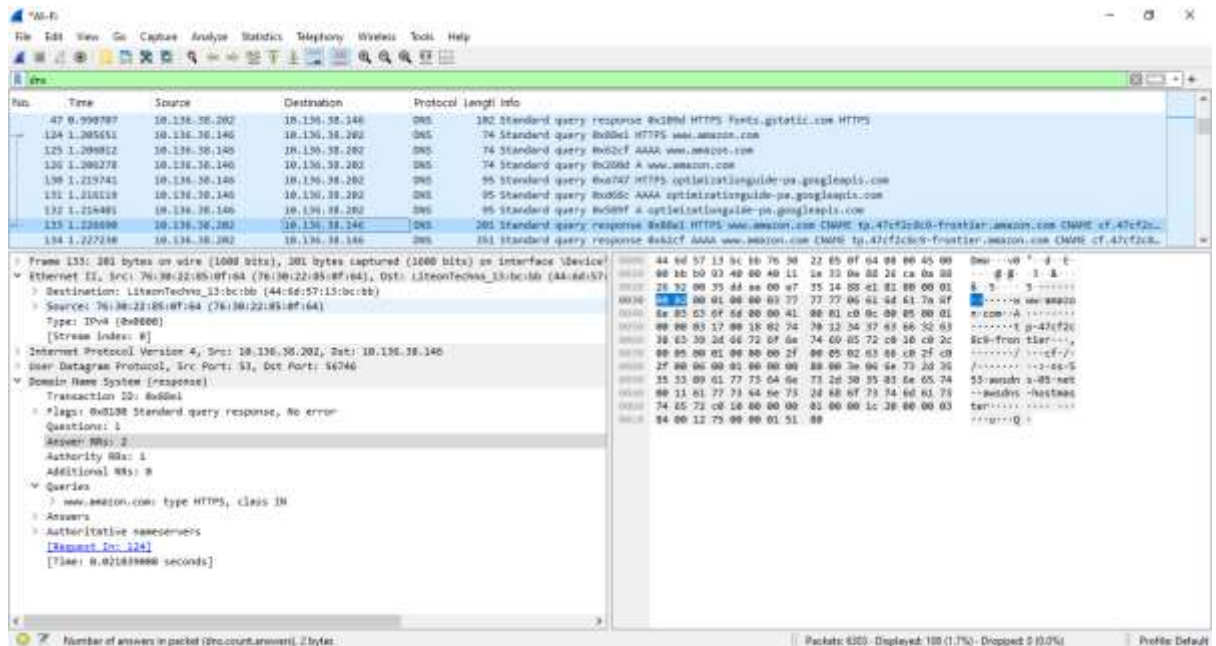
Number of answers in packet (0x00000000): 2 bytes

Packets: 6209 - Displayed: 100 (1.7%) - Dropped: 0 (0.0%)

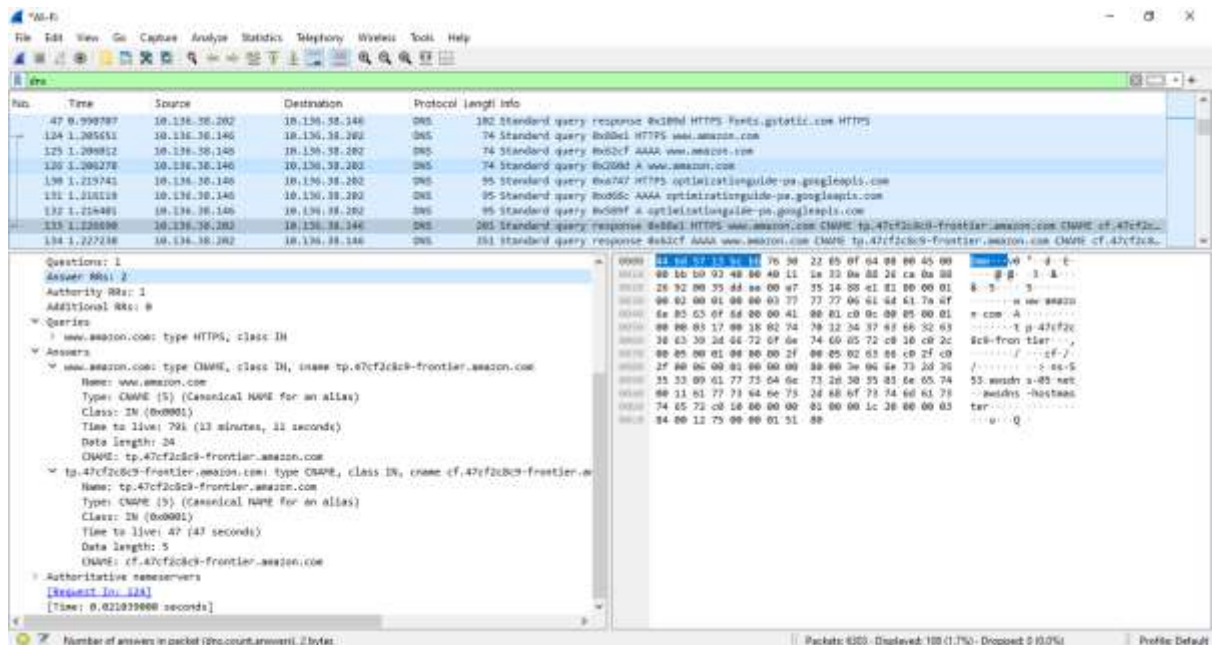
Profile: Default

9. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain?

Ans: The response DNS message contains one answer containing information about the name of the host, the type of address, class, the TTL, the data length.



Extend Answers



10. What Protocol use in Payload? How many bytes use in Payload?

The image shows a Wireshark network traffic capture. The top pane displays a list of network packets. The middle pane shows the details of the selected packet (No. 133), and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
5	0.055889	10.136.38.202	10.136.38.146	DNS	110	Standard query response 0x1b44 AAAA codepen.io AAAA 2000:4790::8810:a329 AAAA 2000:4790::8810:9328
6	0.055881	10.136.38.202	10.136.38.146	DNS	180	Standard query response 0x86cc A codepen.io A 104.16.163.32 A 104.16.147.32
41	0.913357	10.136.38.146	10.136.38.202	DNS	77	Standard query 0x109d HTTPS fonts.gstatic.com
42	0.913383	10.136.38.146	10.136.38.202	DNS	77	Standard query 0x8622 AAAA fonts.gstatic.com
43	0.913391	10.136.38.146	10.136.38.202	DNS	77	Standard query 0x4e97 A fonts.gstatic.com
44	0.913379	10.136.38.202	10.136.38.146	DNS	185	Standard query response 0x8622 AAAA fonts.gstatic.com AAAA 2404:6800:4000:1808::5e
49	0.913434	10.136.38.202	10.136.38.146	DNS	93	Standard query response 0x4e97 A fonts.gstatic.com A 142.251.186.84
47	0.909787	10.136.38.202	10.136.38.146	DNS	180	Standard query response 0x109d HTTPS fonts.gstatic.com HTTPS
124	1.209461	10.136.38.146	10.136.38.202	DNS	74	Standard query 0x88e1 HTTPS www.amazon.com
125	1.209461	10.136.38.146	10.136.38.202	DNS	74	Standard query 0x62bf AAAA www.amazon.com
126	1.209478	10.136.38.146	10.136.38.202	DNS	74	Standard query 0x26bd A www.amazon.com
130	1.215742	10.136.38.146	10.136.38.202	DNS	95	Standard query 0xa747 HTTPS optimizationguide-pa.googleapis.com
131	1.216118	10.136.38.146	10.136.38.202	DNS	95	Standard query 0xb66c AAAA optimizationguide-pa.googleapis.com
132	1.216483	10.136.38.146	10.136.38.202	DNS	95	Standard query 0xc09f A optimizationguide-pa.googleapis.com
133	1.220499	10.136.38.202	10.136.38.146	DNS	260	Standard query response 0x88e1 HTTPS www.amazon.com CNAME tp.47cf2bcb-frontier.amazon.com CNAME cf.47cf2bcb-
134	1.222239	10.136.38.202	10.136.38.146	DNS	351	Standard query response 0x62bf AAAA www.amazon.com CNAME tp.47cf2bcb-frontier.amazon.com CNAME cf.47cf2bcb-
135	1.227651	10.136.38.202	10.136.38.146	DNS	143	Standard query response 0x26bd A www.amazon.com CNAME tp.47cf2bcb-frontier.amazon.com CNAME cf.47cf2bcb-
137	1.229348	10.136.38.202	10.136.38.146	DNS	263	Standard query response 0xa747 HTTPS optimizationguide-pa.googleapis.com CNAME cf.47cf2bcb-

Packet Details (No. 133):

- Frame 133: 260 bytes on wire (2080 bits), 260 bytes captured (2080 bits) on interface libvirt
- Ethernet II, Src: 76:38:32:65:0f:64 (76:38:32:65:0f:64), Dst: LiteonTechno_13:bc:1b (44:6d:15:13:bc:1b)
- Internet Protocol Version 4, Src: 10.136.38.202, Dst: 10.136.38.146
- User Datagram Protocol, Src Port: 53, Dst Port: 53746
 - Source Port: 53
 - Destination Port: 53746
 - Length: 167
 - Checksum: 0x3514 [unverified]
 - [Checksum Status: unverified]
 - [Stream index: 0]
 - [Stream Packet Number: 2]
 - [Timestamps]
 - DNS payload (156 bytes)
- Domain Name System (response)

Raw Data (Hex/ASCII):

```
0000 44 60 57 13 bc 10 76 38 12 15 0f 64 60 60 45 00 0000 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0001 26 32 00 25 4f aa 00 a7 15 34 00 01 01 00 00 01 0001 00 02 00 01 06 00 03 77 77 00 01 5d 61 7a 6f
0002 64 85 55 0f 64 60 60 41 00 01 10 01 00 00 00 01 0002 00 00 03 15 00 13 02 7a 7a 12 34 17 53 04 32 02
0003 18 05 39 20 00 72 0f 00 74 09 05 72 10 38 10 3a 0003 00 05 00 01 00 00 00 2f 00 05 01 03 04 10 2f 10
0004 1f 00 00 00 01 00 00 00 00 00 00 00 3e 00 0a 73 2d 0004 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0005 05 33 09 61 77 73 64 60 73 10 00 33 03 60 05 3a 0005 00 11 11 77 72 64 60 73 2d 00 73 74 6d 01 7a
0006 74 05 72 10 20 00 00 00 00 00 00 00 1c 7a 00 00 00 0006 04 00 12 73 5e 00 01 51 60
```

11. Find **stream index** and **stream packet number** of DNS Response.

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for common actions like opening files, saving, and analyzing. The main window is divided into three panes:

- Packet List:** Shows a single captured packet (No. 45) at Time 0.918434. The packet is a Standard query response from 10.136.38.202 to 10.136.38.146, identified as a GET request to fonts.gstatic.com.
- Packet Details:** Provides a hierarchical view of the packet's structure. It shows the Ethernet II frame, Internet Protocol Version 4 header, and the Hypertext Transfer Protocol (HTTP) request. The HTTP request details include the method (GET), the URL (https://fonts.gstatic.com/), and various headers like User-Agent and Accept.
- Packet Bytes:** Displays the raw data of the packet in hexadecimal and ASCII format.

The status bar at the bottom indicates that 6250 packets are displayed, with 100 (1.7%) dropped and 0.0% of the traffic is profiled.